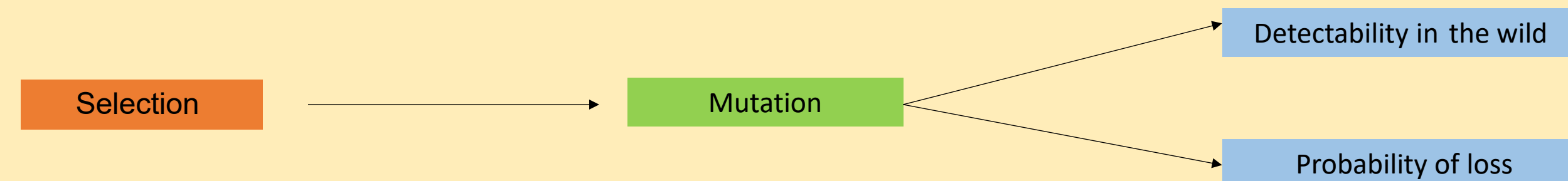


Background & Introduction

- Deleterious mutations cause reduction in effective population size and increase mutational load ultimately lowering the overall fitness and long-term viability of populations. Hence, understanding mutational dynamics and its persistence is important for conservation genetics, particularly in threatened species.
- Detecting deleterious mutations and its trajectory in the wild depends on several interacting factors, including population size (K), selection coefficient (S) & frequency of the mutation (F). However, the combined influence of these parameters on mutation persistence and extinction dynamics remains insufficiently explored.
- In this study, we use forward genetic simulations to investigate how varying K , S , and F shape the fate of deleterious alleles in Olive ridley turtles, a key flagship marine species, classified as vulnerable by the IUCN. This will enable us to optimize sampling protocols for detecting mutation in the wild.

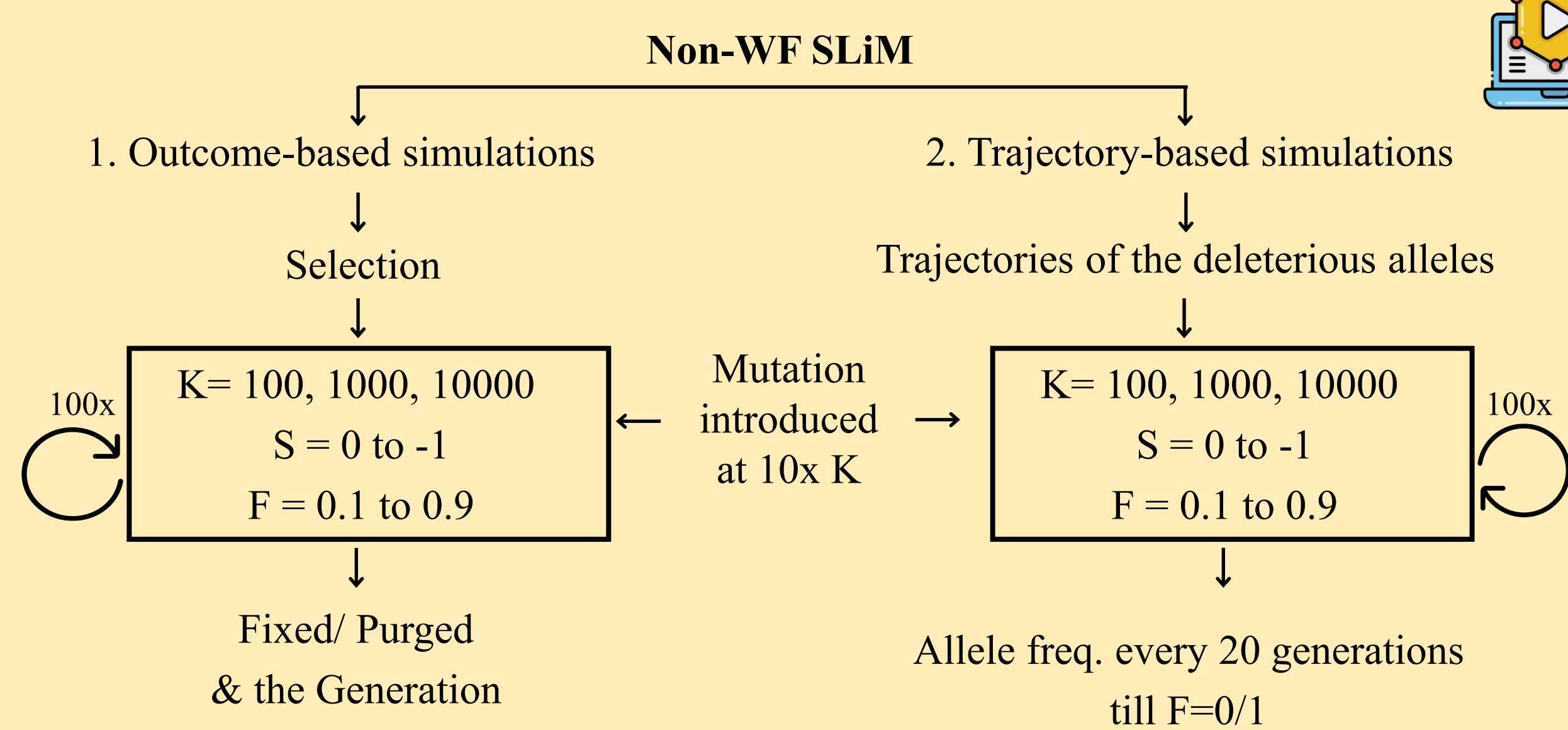


Objectives

- Simulate the probability of loss of a mutant allele under varying combinations of population size (K), selection strength (S) and initial frequency (F).
- Simulate minimum K & S for selection to overcome genetic drift
- Simulate trajectories & detectability of mutation in the wild



Methodology

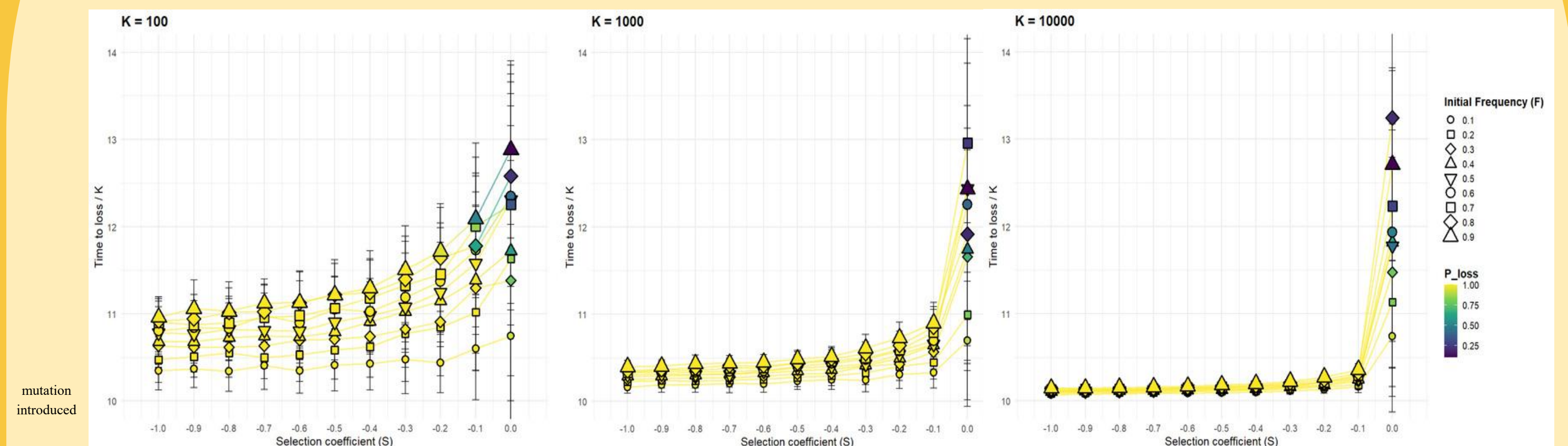


Discussion

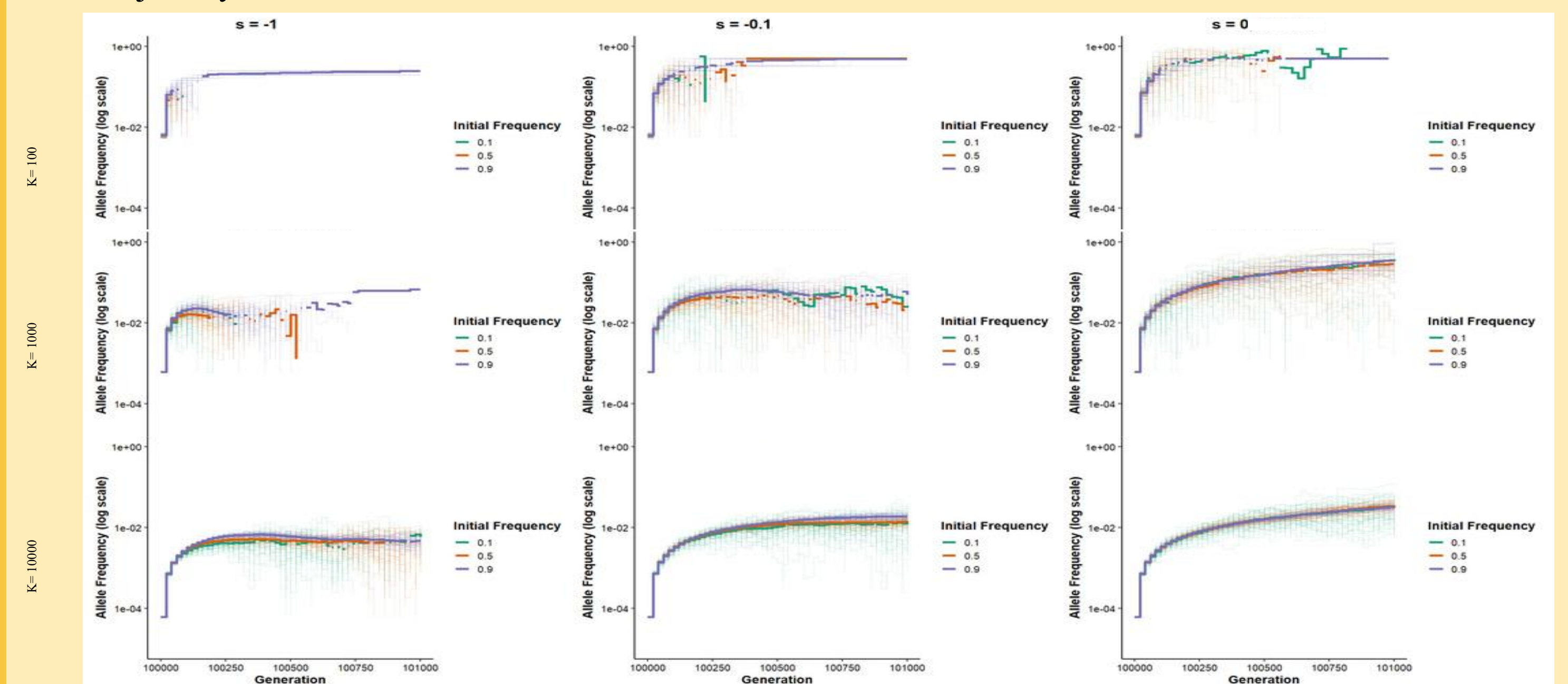
- Strongly deleterious mutations** were **rapidly and consistently purged** across simulations, whereas **weakly deleterious mutations** showed **prolonged persistence** due to the increasing influence of genetic drift.
- Effective population size strongly shaped extinction dynamics: **smaller populations** showed **higher stochasticity**, while **larger populations** allowed **deleterious mutations to persist longer**, especially under weak selection.
- Initial allele frequency significantly affected mutation persistence, with **high-frequency deleterious mutations** taking **longer to be removed** and therefore being more likely to be detected in population samples.
- In **Olive ridley** sea turtle, large population sizes and long generation times may enable weakly deleterious mutations to persist across generations. This signifies the importance of long-term and parameter-aware sampling strategies in conservation genetics.

Results

Time to loss vs S across K & F :



Trajectory Plots:



Summary:

↓ S ↑ Time to loss

↑ F ↓ Time to loss

↑ K ↑ Fluctuation

Future Directions

- Extend these simulations for overlapping generations to simulate realistic turtle population structure.
- Simulate a realistic sampling framework across different effective population size (N_e), sampling ~10 individuals randomly per generation from the first 20 generation and tracking the frequency of mutation to help simulate sampling in the wild.
- Extend simulations to larger population sizes for $K=100,000$ to evaluate how selection operates under these conditions in case of Olive Ridelys in the wild as their population size is ~ 100,000.
- DNA extraction and sequencing of the Olive Ridley samples from last 20 years.
- Identifying deleterious alleles from whole-genome sequencing (WGS) data and assess their distribution and potential impact on survival.
- Field visit to Rushikulya to collect samples during arribada.
- Look for the epigenetic modifications in the genome of Olive Ridelys.
- Compare simulations with empirical data to directly validate whether observed patterns match model predictions

References

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- Ferrera, A., Formia, A., Ciofi, C., Natali, C., Agyekumhene, A., & Allman, P. (2021). Genetic structure of olive ridley sea turtles (*Lepidochelys olivacea*) in Ghana, West Africa. *Journal of experimental marine biology and ecology*, 544, 151614.